

01

Code specifications

Cuts applied in Matteo's original code (per channel): Tauhtauh

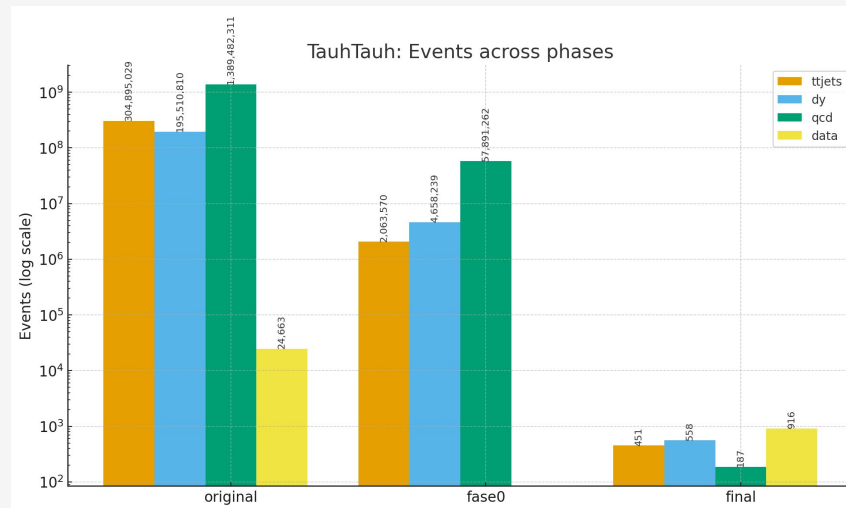
tauhtauh				
	ttjets	dy	qcd	data
FASE0	2 taus	2 taus	2 taus	Not available
Doube_medium	✓	✓	✓	
Iso_Mu24				
Ele32_WpTight				
Electron_				
lumi filtering			✓	
FASE1				
tau id1	>64	>64	>64	>64
tau id2	>8	>8	>8	>8
tau id3	>2	>2	>2	>2
tau pt	>100	>100	>100	>100
mu id				
mu pt				
ele id				
ele pt				
delta R	>0.4	>0.4	>0.4	>0.4
eta	<2.4	<2.3	<2.4	<2.4
charge signs	opposite	opposite	same	opposite
scaling factors		✓		
valid jet and MET info				

no scaling factors
applied in ttjets

Selection cuts

$2 \tau_h$ (ID+ISO)
 $p_T(\tau_h) > 100 \text{ GeV}$
 Opposite sign (same-sign for QCD)
 At least two protons in PPS

Cuts explained in the analysis
note



events surviving per phase in recovered samples

Cuts applied in Matteo's original code (per channel): ETau

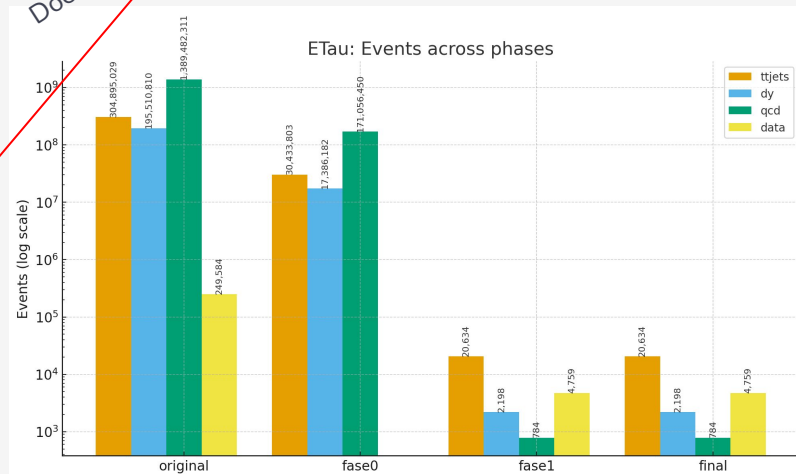
etau			
	ttjets	dy	qcd
FASE0	1 tau, 1 electron	1 tau, 1 electron	1 tau, 1 electron
Doube_medium			
Iso_Mu24			
Ele32_WpTight	✓	✓	✓
Electron_		✓	
lumi filtering	✓		✓
FASE1			
tau id1	>=63	>=63	>64
tau id2	>=7	>=7	>8
tau id3	>=1	>=1	>2
tau pt	>100	>100	>100
mu id			
mu pt			
ele id	1	1	1
ele pt	>35	>35	>35
delta R	>0.4	>0.4	>0.4
eta	<2.4	<2.4	<2.4
charge signs	opposite	opposite	opposite
scaling factors	✓	✓	
valid jet and MET info			

Selection cuts

1 electron and 1 τ_h (ID+ISO)
 $p_T(\text{ele}) > 35 \text{ GeV}$, $p_T(\tau_h) > 100 \text{ GeV}$
 Opposite sign (same-sign for QCD)
 At least two protons in PPS

Cuts explained in the analysis note

Doesn't match!



events surviving per phase in recovered samples

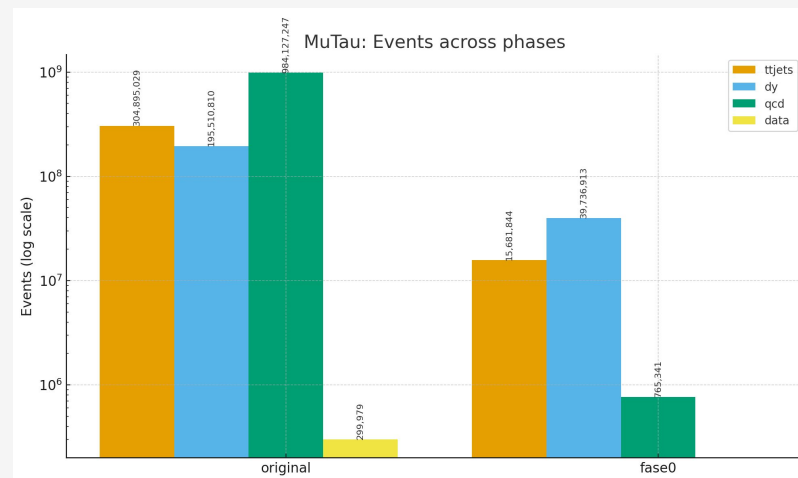
Cuts applied in Matteo's original code (per channel): MuTau

mutau			
	ttjets	dy	qcd
FASE0	1 tau. 1 muon	1 tau. 1 muon	1 tau. 1 muon
Doube_medium			
Iso_Mu24	✓	✓	✓
Ele32_WpTight			
Electron_			
lumi filtering			
FASE1			
tau id1	>63	>63	>63
tau id2	>7	>7	>7
tau id3	>1	>1	>1
tau pt	>100	>100	>100
mu id	>3	>3	>3
mu pt	>35	>35	>35
ele id			
ele pt			
delta R	>0.4	>0.4	>0.4
eta	<2.4	<2.4	<2.4
charge signs	opposite	opposite	same
scaling factors	✓	✓	
valid jet and MET info	✓	✓	

Selection cuts

1 muon and 1 τ_h (ID+ISO)
 $p_T(\mu) > 35 \text{ GeV}, p_T(\tau_h) > 100 \text{ GeV}$
 Opposite sign (same-sign for QCD)
 At least two protons in PPS

Cuts explained in the analysis
note



events surviving per phase in recovered samples

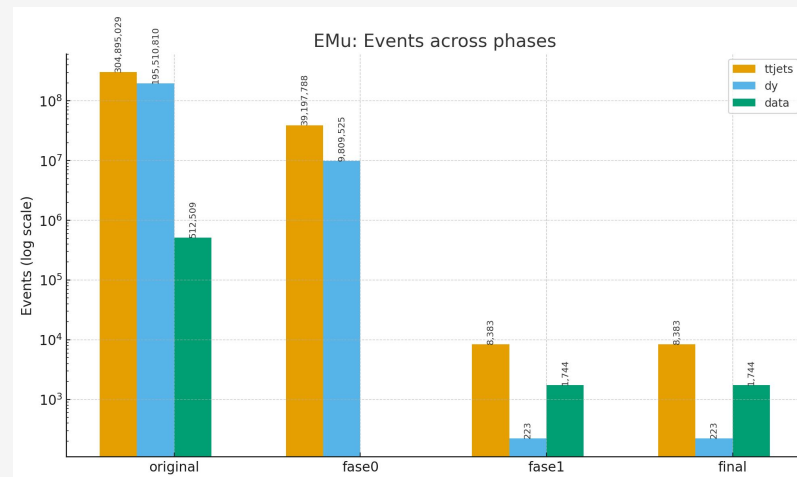
Cuts applied in Matteo's original code (per channel): EMu

emu			
	ttjets	dy	data
FASE0	1 electron, 1 muon	1 electron, 1 muon	1 electron, 1 muon
Doube_medium			✓
Iso_Mu24	✓	✓	
Ele32_WpTight	✓	✓	
Electron_			
lumi filtering	✓		✓
FASE1			
tau id1			
tau id2			
tau id3			
tau pt			
mu id	1	1	1
mu pt	>35	>35	>35
ele id	1	1	1
ele pt	>35	>35	>35
delta R	>0.4	>0.4	>0.4
eta	<2.4	<2.4	<2.4
charge signs	opposite	opposite	opposite
scaling factors	✓	✓	
valid jet and MET info			

Selection cuts

1 electron and 1 μ (ID+ISO)
 $p_T(\mu) > 35 \text{ GeV}, p_T(e) > 35 \text{ GeV}$
 Opposite sign
 At least two protons in PPS

Cuts explained in the analysis
 note



events surviving per phase in recovered samples

Complete cuts applied to each data type - in Matteo's original code- per channel

	ttjets				dy				qcd				data			
	tautau	mutau	etau	emu	tautau	mutau	etau	emu	tautau	mutau	etau	emu	emu	tautauh	etau	mutau
FASE0	2 taus	1 tau. 1 muon	1 tau, 1 electron	1 electron, 1 muon	2 taus	1 tau. 1 muon	1 tau, 1 electron	1 electron, 1 muon	2 taus	1 tau. 1 muon	1 tau, 1 electron	1 electron, 1 muon	2 taus	Not available	Not available	
Doube_medium	✓				✓				✓				✓			
Iso_Mu24		✓		✓		✓		✓		✓						
Ele32_WpTight			✓	✓			✓	✓			✓					
Electron_							✓									
lumi filtering			✓	✓					✓		✓		✓			
FASE1																
tau id1	>64	>63	>=63		>64	>63	>=63		>64	>63	>64			>64		
tau id2	>8	>7	>=7		>8	>7	>=7		>8	>7	>8			>8		
tau id3	>2	>1	>=1		>2	>1	>=1		>2	>1	>2			>2		
tau pt	>100	>100	>100		>100	>100	>100		>100	>100	>100			>100		
mu id		>3				>3				>3				1		
mu pt		>35		>35		>35		>35		>35			>35			
ele id				1				1				1		1		
ele pt			>35	>35			>35	>35			>35		>35			
delta R	>0.4	>0.4	>0.4	>0.4	>0.4	>0.4	>0.4	>0.4	>0.4	>0.4	>0.4		>0.4	>0.4		
eta	<2.4	<2.4	<2.4	<2.4	<2.3	<2.4	<2.4	<2.4	<2.4	<2.4	<2.4		<2.4	<2.4		
charge signs	opposite	opposite	opposite	opposite	opposite	opposite	opposite	opposite	same	same	opposite		opposite	opposite		
scaling factors		✓	✓	✓	✓	✓	✓	✓								
valid jet and MET info		✓				✓										

Number of events per channel - recovered samples

	tjets				dy				qcd			data			
	tauhtau	etau	mutau	emu	tauhtau	etau	mutau	emu	tauhtau	etau	mutau	tauhtau	etau	mutau	emu
SAMPLE TYPE	RunII Summer20UL18NanoAODv9/TTJets_TuneCP5_13TeV				RunII Summer20UL18NanoAODv9/DYJetsToLL_M-50_TuneCP5_13TeV				Run2018/EGamma		Run2018/SingleMuon	Run2018/EGamma		Run2018/SingleMuon	Run2018A/MuonEG
n events	304,895,029				195,510,810				1,389,482,311		984,127,247	1,389,482,311		984,127,247	136,774,293
	tauhtau	etau	mutau	emu	tauhtau	etau	mutau	emu	tauhtau	etau	mutau	tauhtau	etau	mutau	emu
FASE0	2,063,570	30,433,803	15,618,844	39,197,788	4,658,239	17,386,182	39,736,913	9,809,525.00	57,891,262	171,056,450	765,341	24,663	249,584	299,979	512,509
FASE1		20634		8383		2198		223		784			4759		1744
FINAL SAMPLE	451	20,634		8383	558	2198		223	187	784		916			1744

MuTau Channel - comparison in number of events

Available samples:

```
Dados_2018_UL_skimmed_MuTau_nano_fase0_merge.root  QCD_2018_UL_skimmed_MuTau_nano_fase0_merge.root  
DY_2018_UL_skimmed_MuTau_nano_fase0_merge.root    ttJets_2018_UL_skimmed_MuTau_nano_fase0_merge.root
```

MuTau		original samples	my code	original samples	my code	SingleMuon			
	original	304,895,029		195,510,810		984,127,247			
	fase0	15,681,844	42,601,373	39,736,913	42,129,037	765,341	871,204	299,979	
	fase1								
	fase1+pileup protons								

too many events; need to check code

TauTau Channel

Available samples:

Dados_2018_UL_skimmed_TauTau_nano_fase0_march_merge.root	Fundo_syst_2018_UL_skimmed_TauTau_nano_fase1total-protons_2018.root
Dados_2018_UL_skimmed_TauTau_nano_fase1total-protons_2018.root	QCD_2018_UL_skimmed_TauTau_nano_fase0_merge.root
DY_2018_UL_skimmed_TauTau_nano_fase0_merge.root	QCD_2018_UL_skimmed_TauTau_nano_fase1total-protons_2018.root
DY_2018_UL_skimmed_TauTau_nano_fase1total-protons_2018.root	ttJets_2018_UL_skimmed_TauTau_nano_fase0_merge.root
Fundo_2018_UL_skimmed_TauTau_nano_fase1total-protons_2018.root	ttJets_2018_UL_skimmed_TauTau_nano_fase1total-protons_2018.root

ETau Channel

Available samples:

Dados_2018_UL_skimmed_ETau_nano_fase0_merge.root	Fundo_syst_2018_UL_skimmed_ETau_nano_fase1total-protons_2018.root
Dados_2018_UL_skimmed_ETau_nano_fase1total_2018.root	QCD_2018_UL_skimmed_ETau_nano_fase0_merge.root
Dados_2018_UL_skimmed_ETau_nano_fase1total-protons_2018.root	QCD_2018_UL_skimmed_ETau_nano_fase1total_2018.root
DY_2018_UL_skimmed_ETau_nano_fase0_merge.root	QCD_2018_UL_skimmed_ETau_nano_fase1total-protons_2018.root
DY_2018_UL_skimmed_ETau_nano_fase1total_2018.root	ttJets_2018_UL_skimmed_ETau_nano_fase0_merge.root
DY_2018_UL_skimmed_ETau_nano_fase1total-protons_2018.root	ttJets_2018_UL_skimmed_ETau_nano_fase1total_2018.root
Fundo_2018_UL_skimmed_ETau_nano_fase1total-protons_2018.root	ttJets_2018_UL_skimmed_ETau_nano_fase1total-protons_2018.root